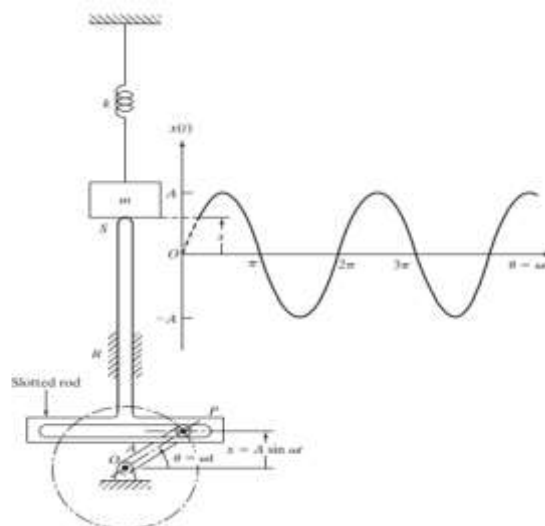


Harmonic motion

The motion of a vibrating system can be **periodic or non periodic**. The best example of a periodic motion is **harmonic motion**.

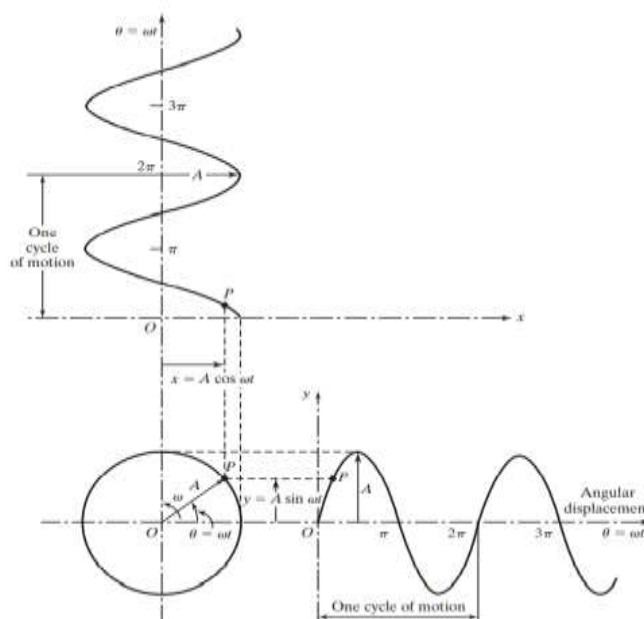
Look at the motion imparted to the mass by the scotch yoke mechanism. The displacement of the mass can be expressed as

$$x = A \sin \omega t$$



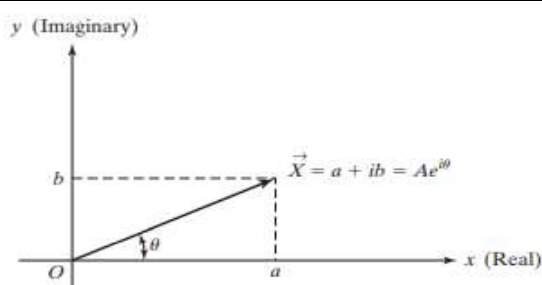
Vectorial representation of harmonic motion

A harmonic motion can be represented by a vector $\vec{X} = \overrightarrow{OP}$ rotating at constant angular velocity ω

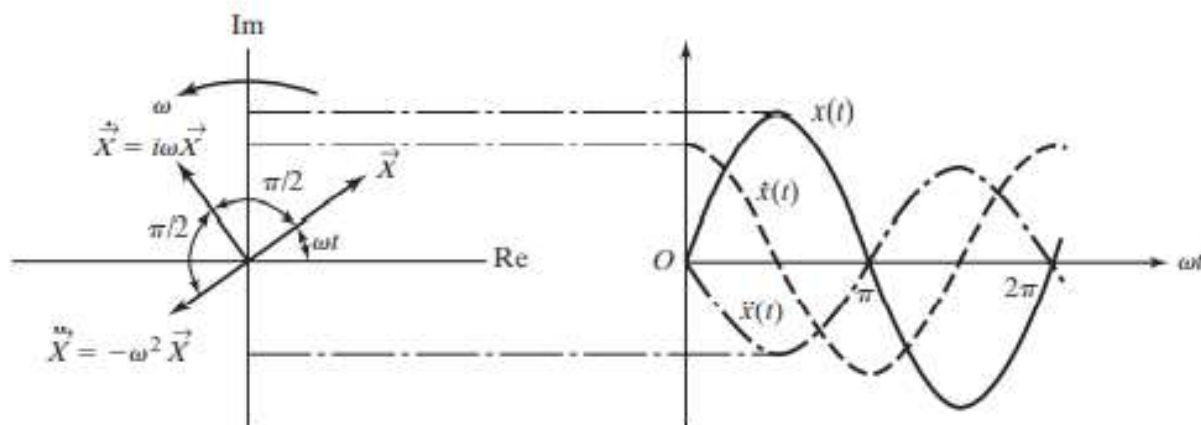


Complex number representation of harmonic function

$$\vec{X} = Ae^{i\theta} = Ae^{i\omega t}$$



Representation of displacement, velocity, and acceleration as rotating vectors



Addition of two harmonic functions

Example 1.

If $x_1 = 2 \sin(2t + 1)$ and $x_2 = 4 \sin(2t + 5)$ find $x(t) = x_1 + x_2$

Let $x(t) = A \sin(2t + \theta)$

$$A \sin(2t + \theta) = 2 \sin(2t + 1) + 4 \sin(2t + 5)$$

$$A \sin 2t \cos \theta + A \cos 2t \sin \theta = 2 \sin 2t \cos 1 + 2 \cos 2t \sin 1 + 4 \sin 2t \cos 5 + 4 \cos 2t \sin 5$$

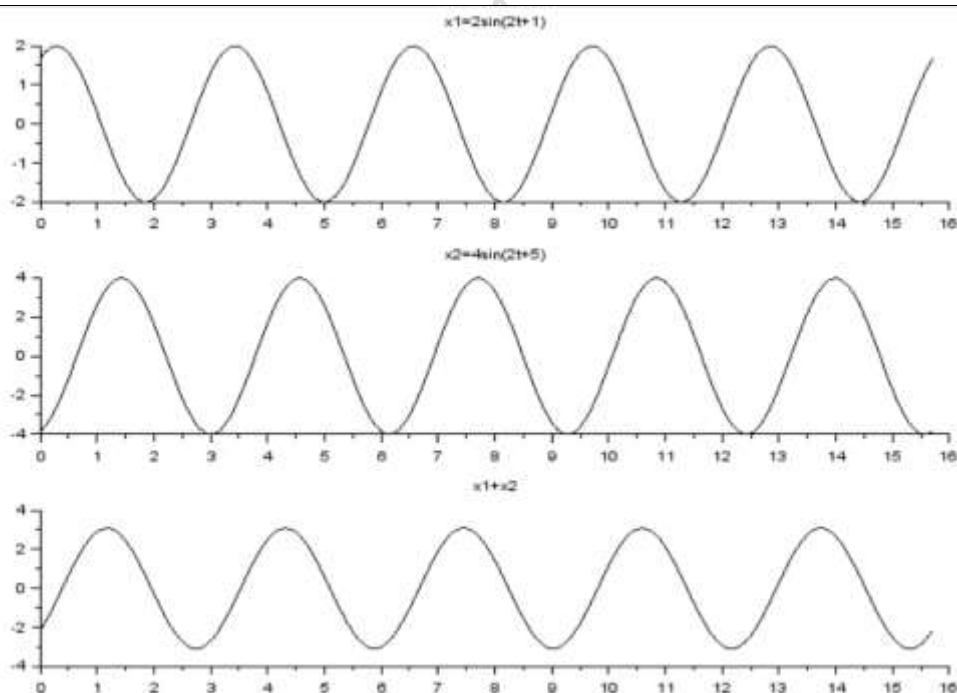
Equating coefficients of $\sin 2t$ and $\cos 2t$

$$A \cos \theta = 2 \cos 1 + 4 \cos 5 = 2.21$$

$$A \sin \theta = 2 \sin 1 + 4 \sin 5 = -2.76$$

$$A = \sqrt{2.21^2 + (-2.76)^2} = 3.13 \quad \tan \theta = \frac{-2.76}{2.21} = -1.25 \quad \ll \gg \quad \theta = -0.89 \text{ rad} = -51.3^\circ$$

$$x(t) = 3.13 \sin(2t - 0.89)$$

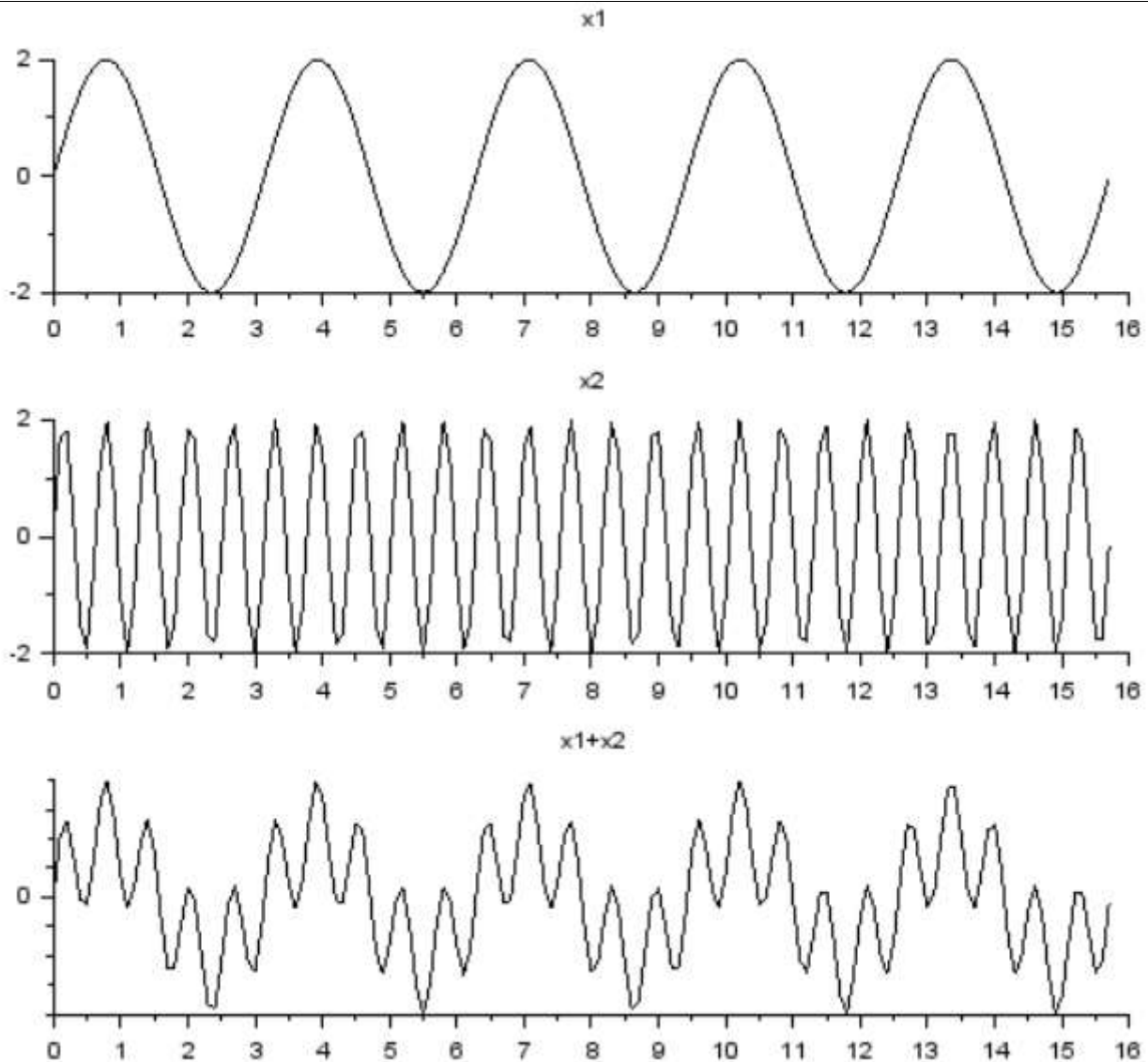


Example 2

$$x_1 = 2 \sin(2t) \quad \ll \gg \quad x_2 = 2 \sin(10t)$$

$$x(t) = x_1 + x_2$$

$$2 \sin(2t) + 2 \sin(10t) = 2(\sin 2t + \sin 10t) = 2 \left(2 \sin \frac{2t + 10t}{2} \cos \frac{10t - 2t}{2} \right) = 4 \cos 4t \sin 6t$$



Example 3

$$x_1 = 4 \sin(10t) \quad \<\> \quad x_2 = 2 \sin(2t)$$

$$x(t) = x_1 + x_2$$

$$4 \sin(10t) + 2 \sin 2t = 4 \sin\left(\frac{10t+2t}{2} + \frac{10t-2t}{2}\right) + 2 \sin\left(\frac{10t+2t}{2} - \frac{10t-2t}{2}\right)$$

$$= 4 \sin(6t + 4t) + 2 \sin(6t - 4t)$$

$$= 4(\sin 6t \cos 4t + \cos 6t \sin 4t) + 2(\sin 6t \cos 4t - \cos 6t \sin 4t)$$

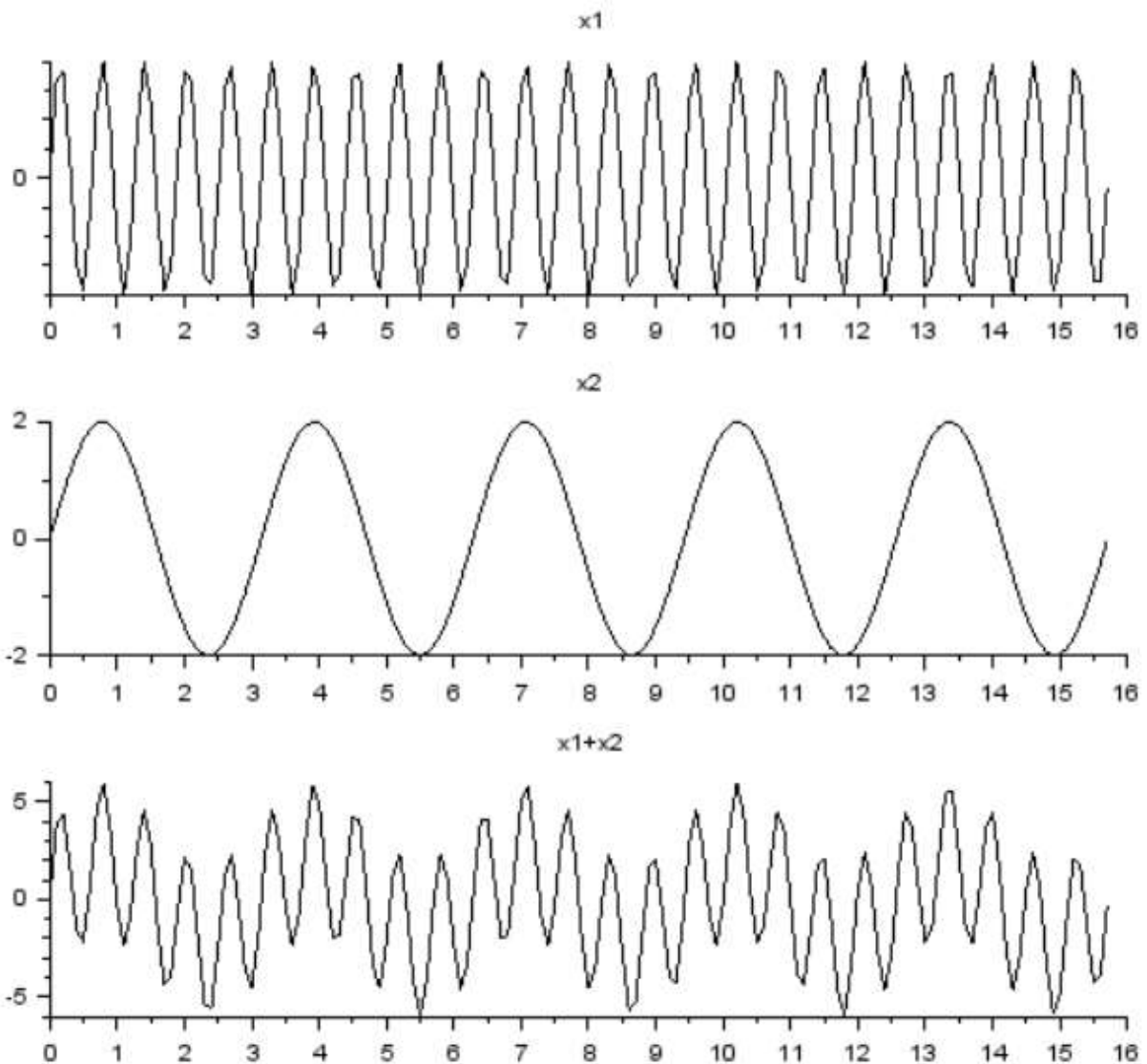
$$= \sin 6t \cos 4t (4 + 2) + \cos 6t \sin 4t (4 - 2)$$

$$A \cos \phi = (4 + 2) \cos 4t \quad \text{and} \quad A \sin \phi = (4 - 2) \sin 4t$$

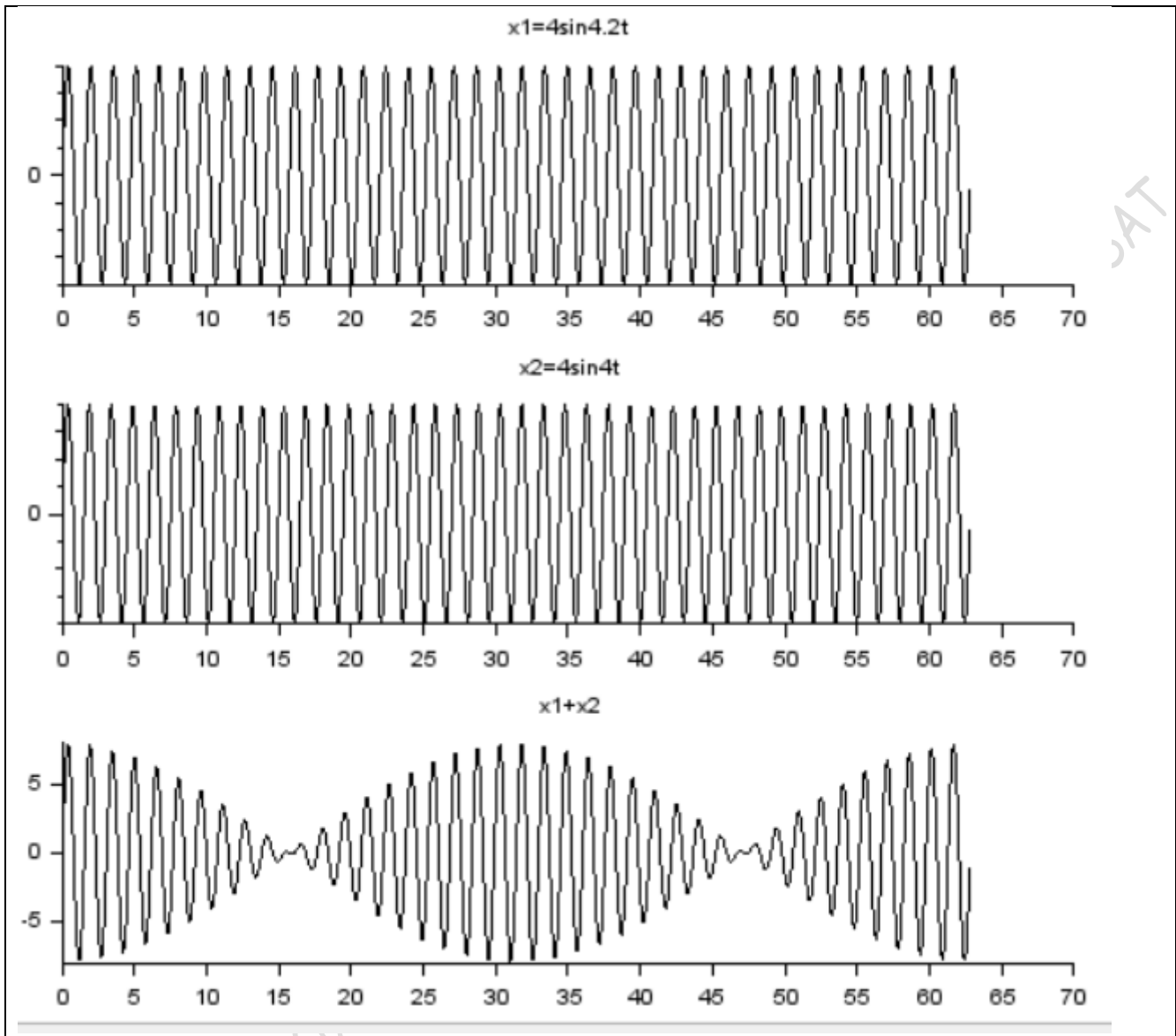
$$\ggg A = \sqrt{\{(4 + 2) \cos 4t\}^2 + \{(4 - 2) \sin 4t\}^2} = \sqrt{4^2 + 2^2 + 2 \times 4 \times 2 \cos 8t} = \sqrt{20 + 16 \cos 8t}$$

$$\tan \phi = \frac{(4 - 2) \sin 4t}{(4 + 2) \cos 4t} = \frac{2}{6} \tan 4t$$

$$x(t) = A \cos \phi \sin 6t + A \sin \phi \cos 6t = A \sin(6t + \phi)$$



When two harmonic signals of slightly different frequencies are superimposed, we get a phenomenon of beats



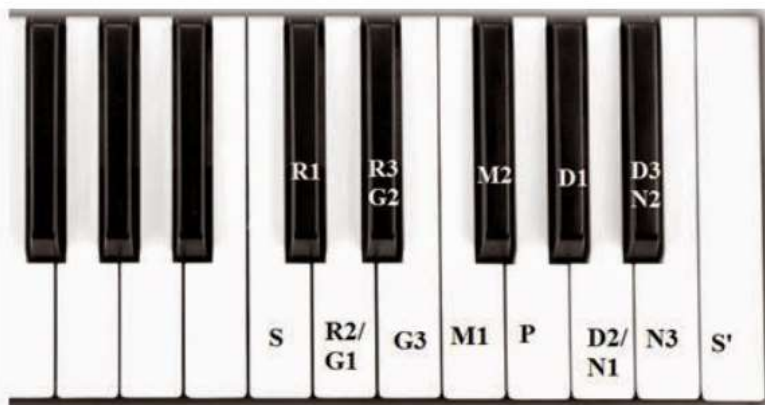
#Exercise

Find the sum of the harmonic functions given below and plot them using Matlab/Scilab.

1. $x_1 = 2 \sin(2t)$, $x_2 = 4 \cos(2t + 5)$
2. $x_1 = 5 \cos(2t - 3)$, $x_2 = 4 \cos(2t + 5)$
3. $x_1 = 10 \sin(5t + 1)$, $x_2 = 10 \sin(5.1t + 2)$
3. $x_1 = 10 \sin(5t + 1)$, $x_2 = 5 \sin(6t + 2)$

OCTAVE

In a frequency band, if the highest frequency is twice the lowest frequency, it is called an **octave**.



Sl. No.	Carnatic Note	Name	Western Note
1	S	Shadjam	C
2	R1	Suddha Rishabham	C#
3	R2/ G1	Chatushruti Rishabham/ Sudha Gandharam	D
4	R3/ G2	Shatshruti Rishabham/ Sadharana Gandharam	D#
5	G3	Antara Gandharam	E
6	M1	Suddha Madhyamam	F
7	M2	Prati Madhyamam	F#
8	P	Panchamam	G
9	D1	Suddha Dhaivatam	G#
10	D2/ N1	Chatushruti Dhaivatam/ Suddha Nishadam	A
11	D3/ N2	Shatshruti Dhaivatam/ Kaisiki Nishadam	A#
12	N3	Kakali Nishadam	B
	S'	(Upper sthayi) Shadjam	C'

A	440
B flat	466
B	494
C	523
C sharp	554
D	587
D sharp	622
E	659
F	698
F sharp	740
G	784
A flat	831
A	880

Decibel

A decibel is originally defined using the ratio of electric powers as

$$dB = 10 \log \left(\frac{P}{P_0} \right)$$

P_0 is some reference value of power. Since electric power is proportional to square of the voltage (amplitude)

$$dB = 10 \log \left(\frac{X}{X_0} \right)^2 = 20 \log \left(\frac{X}{X_0} \right)$$

This equation is used to represent pressure, acceleration, velocity, displacement etc. The reference value for pressure is $20 \mu Pa$ and for acceleration it is $1 \mu g$

Harmonic Analysis

Though harmonic motion is the easiest to handle, many vibration is not harmonic. But we can express any periodic motion as an infinite sum of sine and cosine terms using **Fourier series**.

If $f(t)$ is periodic with period T ,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

ω is the fundamental frequency given by $\frac{2\pi}{T}$

$$a_0 = \frac{2}{T} \int_0^T x(t) dt \quad \langle\langle\rangle\rangle \quad a_n = \frac{2}{T} \int_0^T x(t) \cos n\omega t dt \quad \langle\langle\rangle\rangle \quad b_n = \frac{2}{T} \int_0^T x(t) \sin n\omega t dt$$

This can be also written as a series involving *sine* terms only or *cosine* terms only

For example,

by substituting $a_n = d_n \sin \varphi_n$ and $b_n = d_n \cos \varphi_n$, where $d_n = \sqrt{a_n^2 + b_n^2}$ and $\tan \varphi_n = \frac{a_n}{b_n}$

$$a_1 \cos \omega t + b_1 \sin \omega t = d_1 \sin \varphi_1 \cos \omega t + d_1 \cos \varphi_1 \sin \omega t = d_1 \sin(\omega t + \varphi_1)$$

$$f(t) = d_0 + \sum_{n=1}^{\infty} d_n \sin(n\omega t + \varphi_n)$$

Complex form of the Fourier series

$$\cos n\omega t = \frac{e^{in\omega t} + e^{-in\omega t}}{2} \quad \langle\langle\rangle\rangle \quad \sin n\omega t = \frac{e^{in\omega t} - e^{-in\omega t}}{2i}$$

$$f(t) = \left(\frac{a_0 - ib_0}{2}\right) e^{i(0)\omega t} + \sum_{n=1}^{\infty} a_n \left(\frac{e^{in\omega t} + e^{-in\omega t}}{2}\right) + b_n \left(\frac{e^{in\omega t} - e^{-in\omega t}}{2i}\right) \quad \text{where } b_0 = 0$$

$$f(t) = \sum_{n=0}^{\infty} \left(\frac{a_n - ib_n}{2}\right) e^{in\omega t} + \left(\frac{a_n + ib_n}{2}\right) e^{-in\omega t}$$

$$f(t) = \sum_{-\infty}^{\infty} c_n e^{in\omega t} \quad \langle\langle\rangle\rangle \quad \left(\frac{a_n - ib_n}{2}\right) = c_n \quad \langle\langle\rangle\rangle \quad \left(\frac{a_n + ib_n}{2}\right) = c_{-n}$$

$$c_n = \left(\frac{a_n - ib_n}{2}\right) = \frac{1}{T} \int_0^T x(t) [\cos n\omega t - i \sin n\omega t] dt = \frac{1}{T} \int_0^T x(t) e^{-in\omega t} dt$$

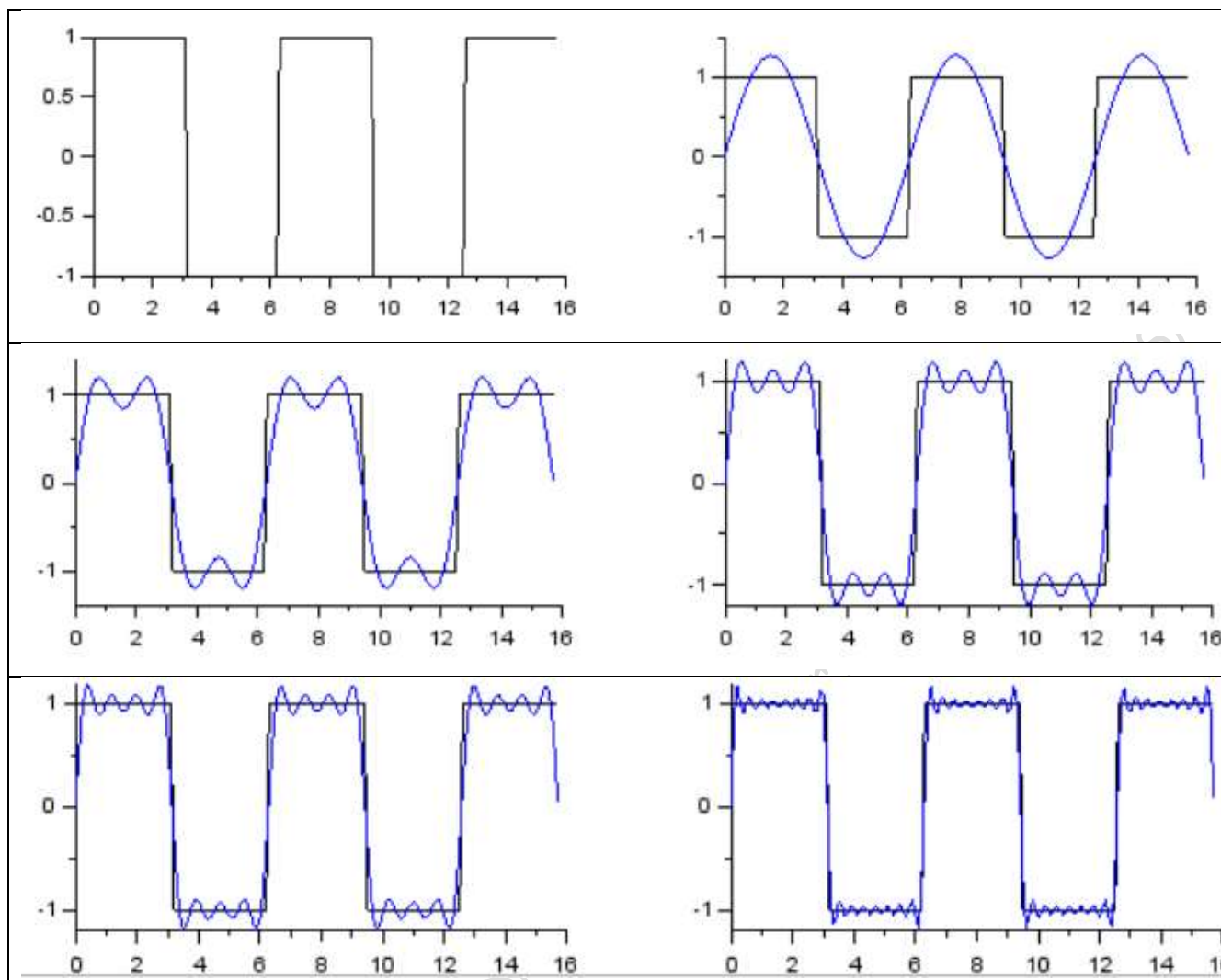
For example, consider a square wave

$$f(t) = 1, \quad 0 < t < \pi \\ = -1, \quad \pi < t < 2\pi$$

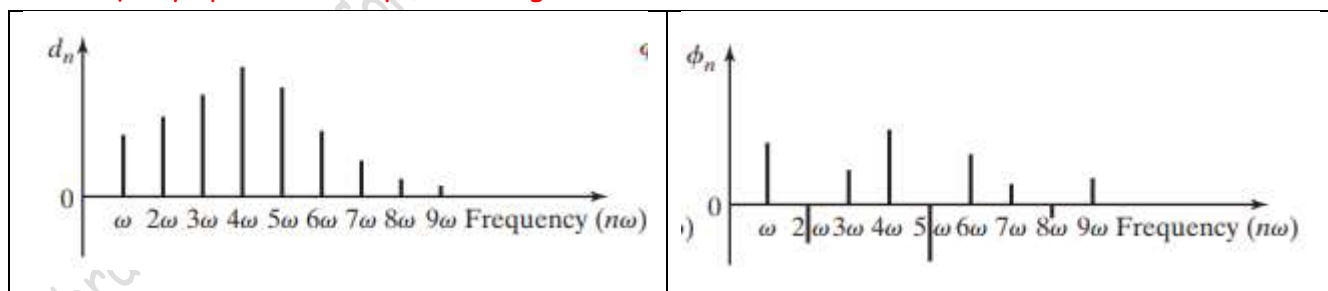
The fourier series expansion for the above is given by

$$f(t) = \sum_{n=1,3,5}^{\infty} \frac{4}{n\pi} \sin nt = \frac{4}{\pi} \sin t + \frac{4}{3\pi} \sin 3t + \frac{4}{5\pi} \sin 5t + \dots \dots \dots$$

The one term, two term, three term, for term and seven term approximations are shown below.

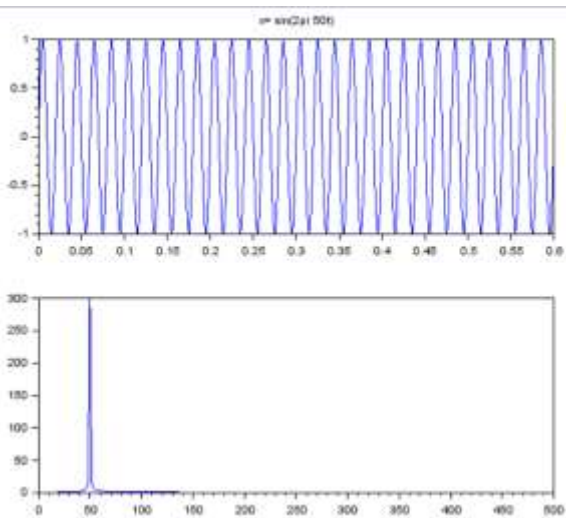


If the coefficients a_n and b_n or d_n and ϕ_n are plotted against the frequencies $n\omega$, it is known as the **frequency spectrum** or **spectral diagram**.

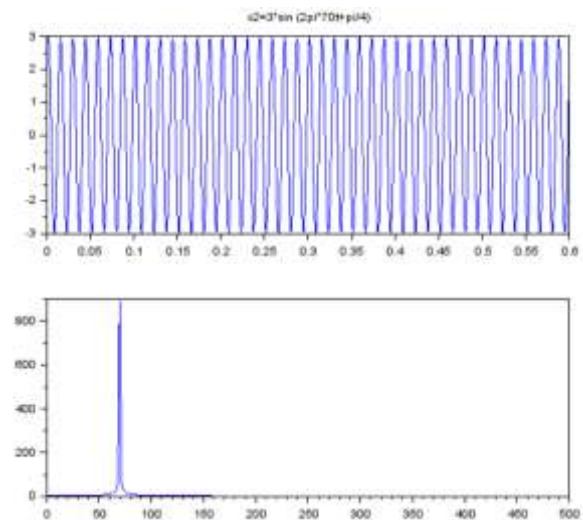


Fast Fourier Transform (FFT)

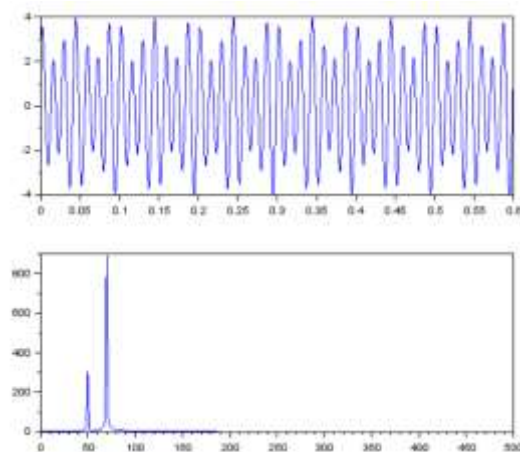
$$x_1 = \sin(2\pi \times 50 \times t)$$



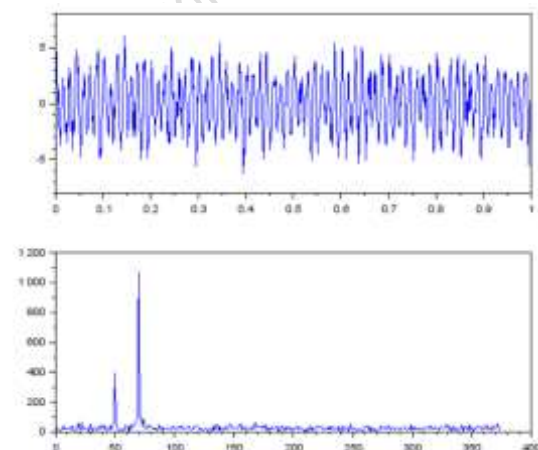
$$x_2 = 3 \sin\left(2\pi \times 70 \times t + \frac{\pi}{4}\right)$$



$$x_3 = \sin(2\pi \times 50 \times t) + 3 \sin\left(2\pi \times 70 \times t + \frac{\pi}{4}\right)$$



$$x_3 = \sin(2\pi \times 50 \times t) + 3 \sin\left(2\pi \times 70 \times t + \frac{\pi}{4}\right) + \text{random noise}$$



Exercise 1

If the Fourier series expansion of the motion of the valve is given by

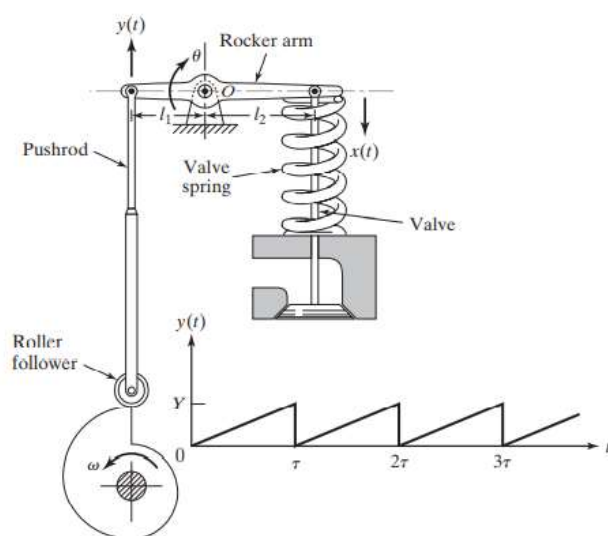
$$x(t) = \frac{At}{\tau}$$

$$= \frac{2}{\pi} \left\{ \frac{\pi}{2} - \left[\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \dots \right] \right\}$$

Plot the one term, two terms, three terms, four terms approximations using Mtlab or Scilab.

$$\text{Take } A = Y \frac{l_2}{l_1} = 2,$$

$$\omega = 2 \quad \text{and} \quad \tau = \frac{2\pi}{\omega}$$



Matlab/Scilab Assignment

1. Considering the sampling frequency 1 000Hz and a signal duration 1.5 seconds, plot the sinusoidal signals of
 - (a) Amplitude 0.8 and frequency 30 Hz
 - (b) Amplitude 0.35 and frequency 50 Hz
 - (c) Amplitude 1 and frequency 120 Hz
 - (d) $S=a+b+c$
 - (e) Corrupt the signal 'S' with zero-mean white noise with a variance of 4
 Compute the Fourier transform of the signal and plot the 2 sided spectrum and single sided spectrum in each case. Show the influence of sampling frequency on the results by changing it to 500 Hz and 100 Hz.
2. Plot a square wave of amplitude 1 and frequency 5 Hz for 2 seconds. Plot its FFT taking a sampling frequency of 150 Hz.
3. Plot a rectangular pulse of amplitude 1 and frequency 5 Hz for 2 seconds. Plot its FFT taking a sampling frequency of 150 Hz.
4. Consider audio data collected from underwater microphones off the coast of California. This data can be found in a library maintained by the Cornell University Bioacoustics Research Program. Load and format a subset of the data in bluewhale.au, which contains a Pacific blue whale vocalization. You can use the command `sound(x,fs)` to listen to the entire audio file .
(Ref:: <https://in.mathworks.com/help/matlab/math/fourier-transforms.html>)

Plot the signal using the following commands.

```
whaleFile = 'bluewhale.au';
[x,fs] = audioread(whaleFile);
whaleMoan = x(2.45e4:3.10e4);
t = 10*(0:1/fs:(length(whaleMoan)-1)/fs);
plot(t,whaleMoan)
xlabel('Time (seconds)')
ylabel('Amplitude')
xlim([0 t(end)])
```

Plot the power spectrum

5. Create your own audio file that is at least 5 seconds long. Use the `waverecord` command in MATLAB and a sampling frequency of 10 kHz. Use the `sound` command in MATLAB to play the sound.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

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